PREHOSPITAL STROKE PROTOCOL

Comprehensive Stroke Center (CSC) Referral Network

Aligned with the 2026 AHA/ASA Guideline for the Early Management of Patients with Acute Ischemic Stroke

Document Status - DRAFT - for internal stroke committee review - Source: Prabhakaran S, et al. 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke. Stroke. 2026 (replaces the 2018 guideline and 2019 update). - This summary must be cross-checked against state/regional EMS scope-of-practice regulations before adoption.

1. Dispatch & Public Awareness

[1] Sustained public education programs on stroke recognition (FAST/BE-FAST) and the need to call 9-1-1, targeted to diverse communities and reinforced over time.

[2a] EMS dispatchers should use a validated telephone stroke assessment tool to support early identification and prioritize dispatch.

[1] Targeted stroke education for EMS professionals and physicians to reduce prehospital delay and maximize treatment eligibility.

2. On-Scene Assessment

2.1 Stroke Identification

[1] Use a brief, validated stroke screening tool (e.g., Cincinnati Prehospital Stroke Scale) for all patients with suspected stroke.

[1] Apply a validated LVO severity scale (RACE, VAN, LAMS, or FAST-ED) to identify candidates for direct thrombectomy-center transport.

2.2 Core Data to Capture

Last known well (LKW) time - establish precisely; drives the entire treatment window

Point-of-care blood glucose (rule out hypo/hyperglycemia mimics)

Baseline vital signs, including SpO?

Focused history: anticoagulant/antiplatelet use, recent surgery or trauma, seizure at onset

Witness/bystander contact information for onset time verification

3. Field Interventions

3.1 Recommended

[1] Airway support and ventilatory assistance as needed for decreased consciousness or bulbar dysfunction.

[1] Supplemental oxygen only if hypoxic, titrated to SpO? > 94%.

[1] Correct hypotension/hypovolemia to maintain systemic perfusion.

3.2 Do Not Perform (Class 3 - No Benefit / Harm)

AVOID - Remote ischemic conditioning via arm BP cuff inflation - no functional benefit.

AVOID - Prehospital transdermal glyceryl trinitrate (nitroglycerin) - potentially harmful.

AVOID - Intensive field BP lowering to 130-140 mmHg systolic - no functional benefit.

AVOID - Routine supplemental oxygen in non-hypoxic patients - no benefit.

4. Advance Notification

[1] Provide advance notification to the receiving hospital for every suspected stroke transport - reduces in-hospital evaluation time, increases thrombolytic use, and decreases mortality.

Notification should include: LKW time, stroke scale/LVO severity score, vitals, glucose, anticoagulant status, and estimated time of arrival, communicated per regional stroke alert protocol.

5. EMS Destination Decision-Making

[1] Transport to the closest appropriate stroke-capable facility (ASRH, PSC, TSC, or CSC) rather than a non-stroke-capable hospital.

5.1 Areas with Local Access to a Thrombectomy-Capable Center

[2a] For suspected LVO stroke, direct transport to a thrombectomy-capable center (TSC/CSC) over initial transport to a non-TSC with secondary transfer.

5.2 Areas with a Well-Coordinated Stroke System of Care

AVOID - Bypassing a local, thrombolysis-capable stroke center for a distant (45-60 min) thrombectomy center does not improve 3-month outcomes where transfer networks are reliable - "drip-and-ship" remains appropriate.

5.3 Areas Without a Well-Coordinated Stroke System

[2b] Direct transport of suspected LVO patients to the closest thrombectomy-capable center may be reasonable, provided it will not disqualify the patient from IV thrombolysis.

5.4 Interhospital Transfer

[1] pre-established transfer agreements and protocols between spoke facilities and the CSC to minimize door-in-door-out (DIDO) time.

6. Mobile Stroke Units (If Deployed in Network)

[1] Use MSUs over conventional EMS, where available, for thrombolytic-eligible patients to minimize onset-to-treatment time.

[1] MSUs must be equipped to diagnose and administer IV thrombolysis in the field.

[1] MSU care should include streamlined protocols and neurological expertise (in-person or telemedicine).

[2a] For EVT-eligible patients, MSUs can help identify and triage to the correct thrombectomy-capable facility with prehospital notification of the stroke team.

7. Prehospital Telemedicine

[1] Consider in-ambulance telemedicine, when feasible, to complement paramedic assessment and identify candidates for reperfusion therapy.

8. CSC Arrival & Handoff

Stroke team activated in advance based on prehospital notification (physician, nursing, imaging, lab)

NIHSS performed immediately on arrival

Emergent non-contrast CT +/- CTA - target completion within 25 minutes of arrival

Confirm point-of-care glucose

Consider direct-to-angiography-suite pathway for LVO patients meeting severity criteria (e.g., RACE > 4) and EVT eligibility

Telestroke/teleradiology support available 24/7 for imaging review and treatment decision-making at all network sites

9. Time-to-Treatment Targets

Metric	Target
Door-to-CT (imaging) completed	? 25 minutes
Door-to-needle (IVT)	? 60 minutes (? 45 min goal for CSC)
Door-to-puncture (EVT groin/access)	As fast as achievable - tracked per case
Door-in-door-out (transferring facility)	? 60 minutes
EMS on-scene time (suspected stroke)	? 15 minutes when feasible

10. Quality Improvement & Metrics

[1] Track door-to-puncture time, successful reperfusion (mTICI ?2b), and long-term patient outcomes for all EVT cases.

[1] Credential neurointerventionalists against established, agreed-upon training and certification standards.

[1] Participate in a stroke data registry and engage in continuous, multicomponent QI review to reduce disparities and improve outcomes.

Recommendation Class Legend

[1] Strong (benefit >>> risk) [2a] Moderate (benefit >> risk) [2b] Weak (benefit ? risk)

[3H] Harm - should not be performed [3N] No benefit - not recommended

Source: Prabhakaran S, et al. 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke: A Guideline From the American Heart Association/American Stroke Association. Stroke. 2026;10.1161/STR.0000000000000513. Replaces the 2018 AIS Guideline and 2019 update.